

Topic 2

Supervised Machine Learning- Regression

Topics

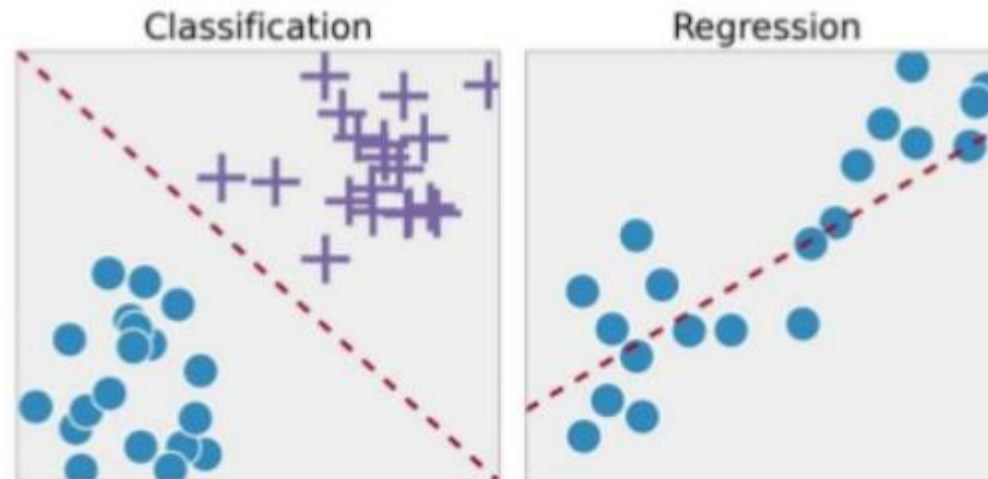
- Introduction to Machine Learning
- Overview of Supervised Learning - Regression
- Linear Regression
- Polynomial Regression
- Support Vector Machines

Supervised learning

In supervised learning, we are given a data set and already know what is the correct output based on the given input. Supervised learning will learn a pattern to represent the relationship between the input and output.

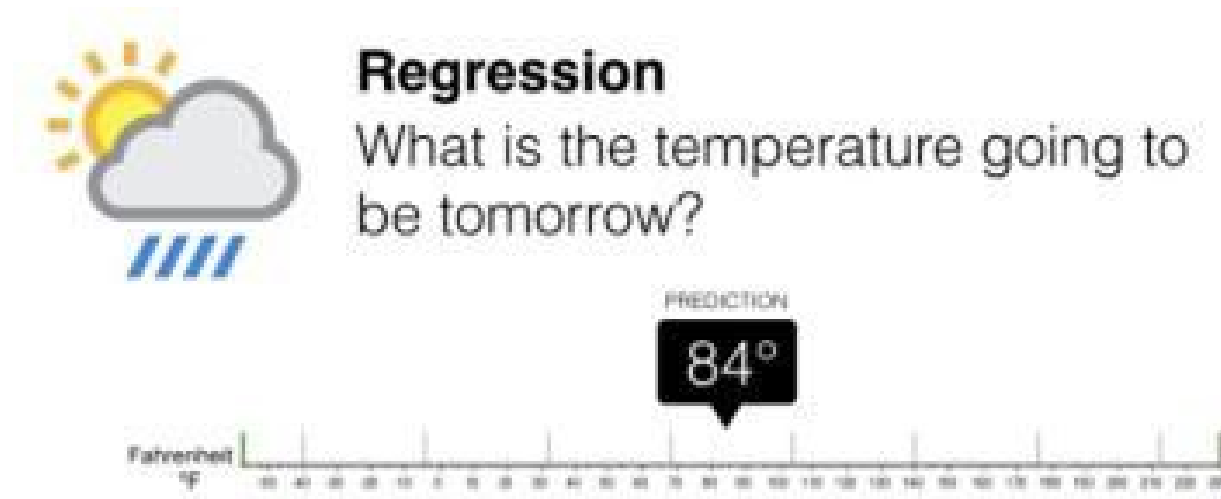
Two types of Supervised Learning :-

1. Regression — Estimate continuous values (Real valued output)
2. Classification — Identify a unique class (Discrete values, Boolean or Categories)



Supervised learning - Regression

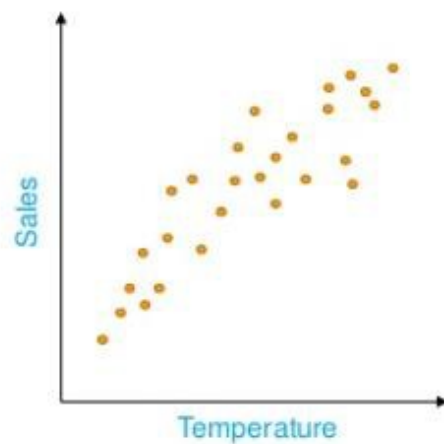
Regression models give a target prediction value based on independent variables. It is mostly used for finding out the **relationship between variables** and **forecasting**. Regression can be used to estimate/ predict **continuous values** (Real valued output).



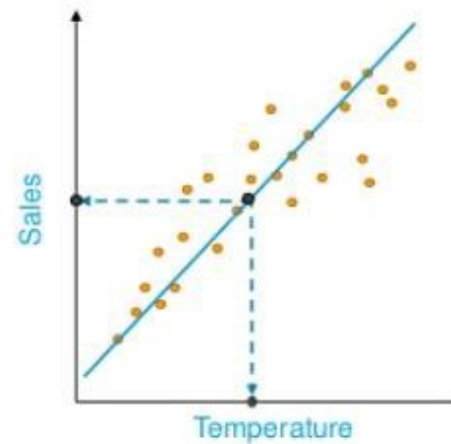
Supervised learning-Linear Regression

Linear Regression is a linear modelling algorithm to find relationship between one or more independent variables denoted as X and dependent variable (target) denoted as Y

Example: Regression Prediction of sales of Ice Cream based on temperature



Plotting the sales of ice cream based on temperature



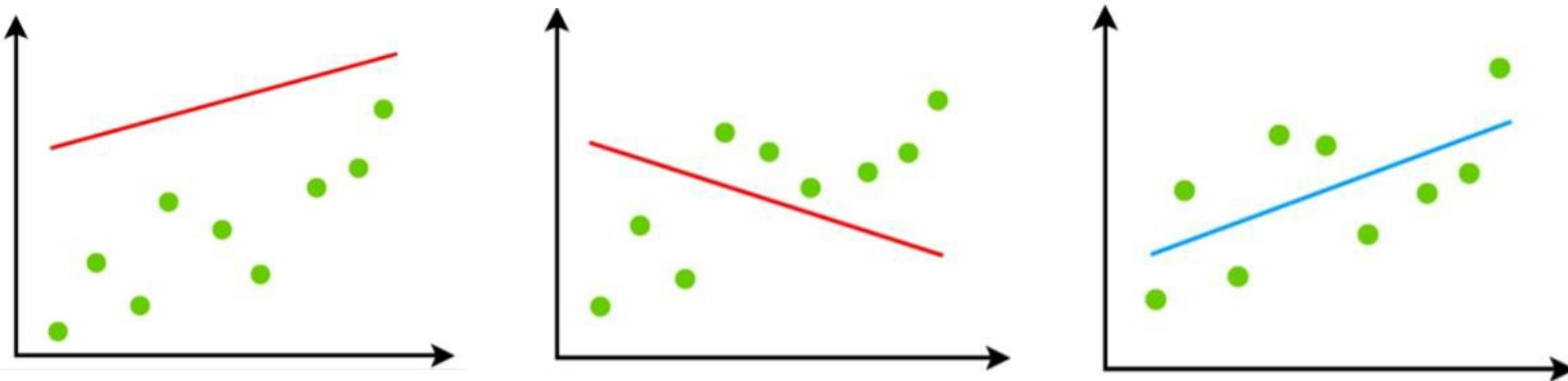
Regression line to predict the sales

$$y = wx + b$$

Supervised learning-Linear Regression

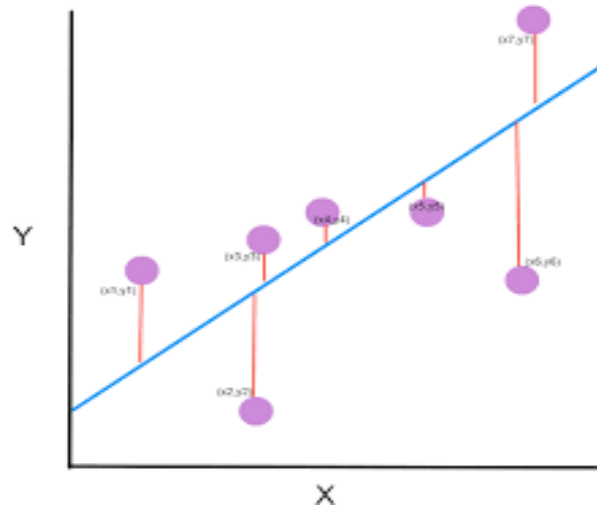
Linear Regression approach

Finding the best fit line: The best fit line can be found by minimizing the distance between all the data points and the distance to the regression line. Ways to minimize this distance are sum of squared errors, sum of absolute errors etc.



Supervised Machine Learning-Linear Regression

Regression performance measurement
Mean Absolute Error.

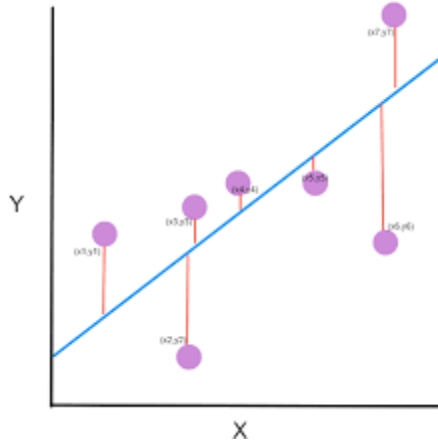


Mean Absolute Error

$$\text{MAE} = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|$$

Supervised learning-Linear Regression

Regression performance measurement
Mean Square Error.



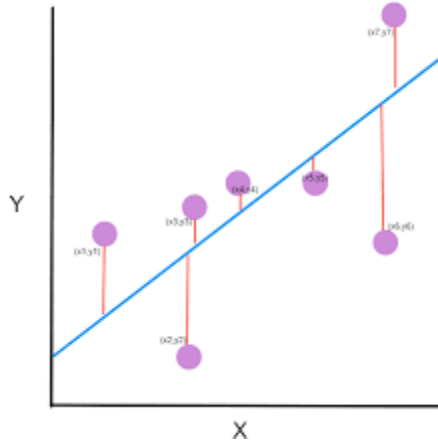
Mean Square Error

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

Supervised learning-Linear Regression

Regression performance measurement

Root Mean Square Error



Root Mean Square Error

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2}$$

Supervised learning-Linear Regression

Regression performance measurement

- The **correlation**, denoted by r , measures the amount of linear association between two variables.
- r is always between -1 and 1 inclusive.
- The **R-squared value(Coefficient of Determination)**, denoted by R^2 , is the square of the correlation. It measures the proportion of variation in the dependent variable that can be attributed to the multiple independent variables.
- The R-squared value R^2 is always between 0 and 1 inclusive.

Supervised learning-Linear Regression

R-squared value

- Perfect positive linear association. The points are exactly on the trend line.

Correlation $r = 1$; R-squared = 1.00

- Large positive linear association. The points are close to the linear trend line.

Correlation $r = 0.9$; R-squared = 0.81.

- Small positive linear association. The points are far from the trend line.

Correlation $r = 0.45$; R-squared = 0.2025.

Supervised learning-Linear Regression

R-squared value

- No association. There is no association between the variables.
Correlation $r = 0.0$; R-squared = 0.0.
- Small negative association.
Correlation $r = -0.3$. R-squared = 0.09.
- Large negative association.
Correlation $r = -0.95$; R-squared = 0.9025
- Perfect negative association.
Correlation $r = -1$. R-squared = 1.00.

Supervised learning-Linear Regression

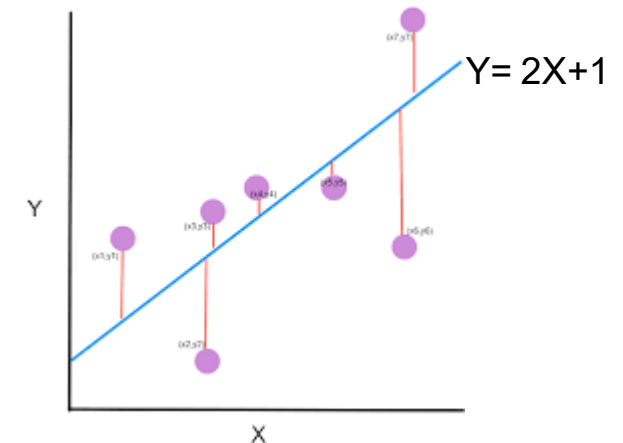
Training process for the model $Y = WX + B$

Given X and Y column know data

This is the training process to get the result of $W=2$ and $B=1$

$$Y = 2X + 1$$

Feature	Label	Adjust W/B		Predict	Error=Y-Yhat
X	Y	W	B	Yhat	error
1	3	0	0	0	3
2	5	0.5	0.1	1.1	3.9
3	7	0.9	0.5	3.2	3.8
4	9	1	0.7	4.7	4.3
5	11	2	2	12	-1
6	13	2.1	1	13.6	-0.6
7	15	2	1	15	0



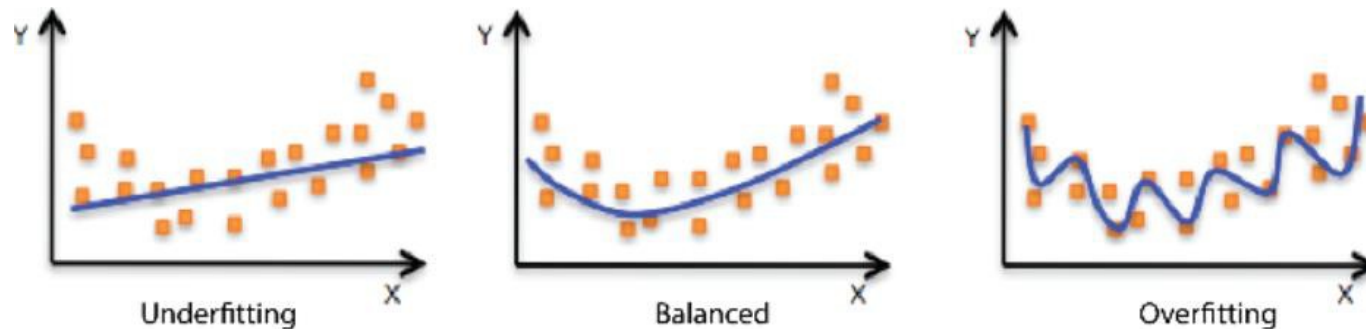
Actual Value	Predicted Value	Residuals	MAE	RMSE
[1.5, 1.8, 2, 2.5, 3]	[1.1, 2, 1.7, 2, 2.8]	[0.4, 0.2, 0.3, 0.5, 0.2]	0.32	0.34
[1.5, 1.8, 2, 2.5, 3]	[1.1, 2, 1.7, 2, 1]	[0.4, 0.2, 0.3, 0.5, 2]	0.68	0.95
[1.5, 1.8, 2, 2.5, 3]	[1.1, 1.4, 1.6, 2.1, 2.6]	[0.4, 0.4, 0.4, 0.4, 0.4]	0.4	0.4

RMSE is sensitive to error distribution of samples

Supervised learning-Linear Regression

Model Fit: Underfitting vs. Overfitting

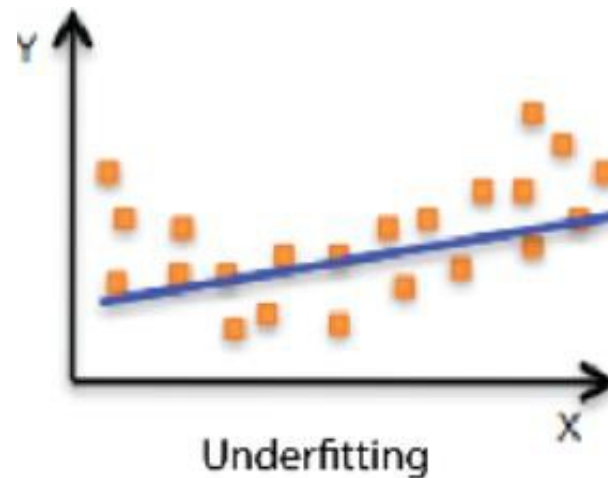
Understanding model fit is important for understanding the root cause for poor model accuracy. This understanding will guide you to take corrective steps. We can determine whether a predictive model is underfitting or overfitting the training data by looking at the prediction error on the training data and the evaluation data.



Supervised learning-Linear Regression

Model Fit: Underfitting

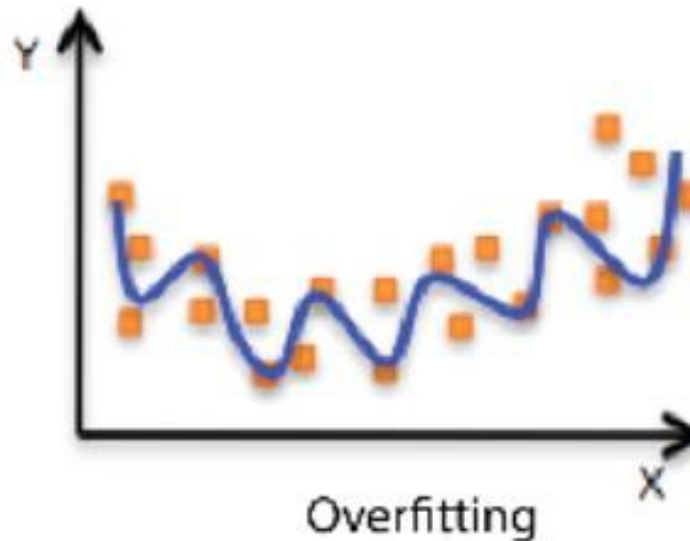
Your model is *underfitting* the training data when the model performs poorly on the training data. This is because the model is unable to capture the relationship between the input examples (often called X) and the target values (often called Y).



Supervised learning-Linear Regression

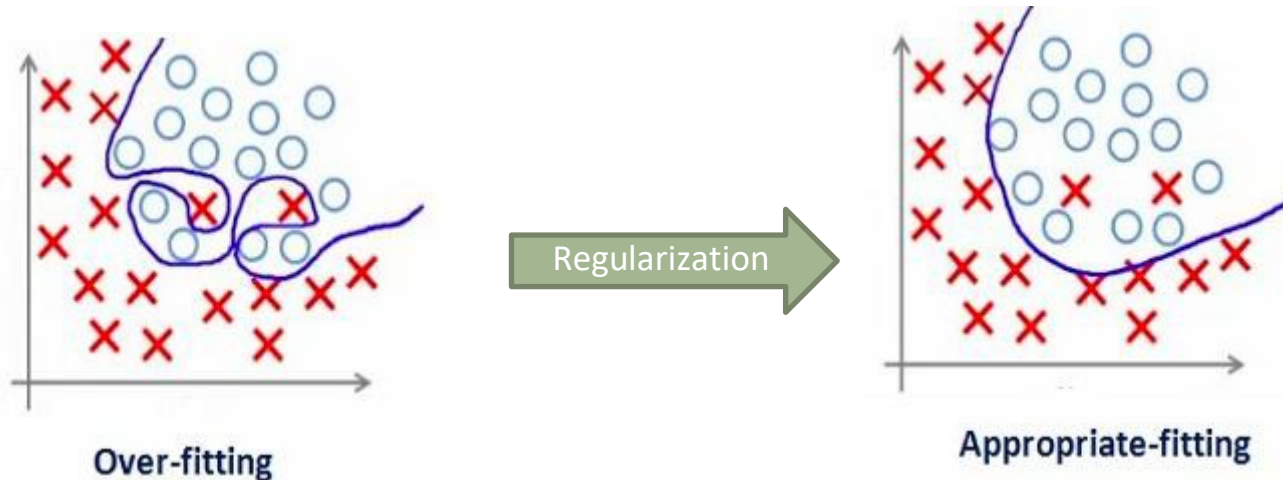
Model Fit: Overfitting

Your model is *overfitting* your training data when you see that the model performs well on the training data but does not perform well on the evaluation data. This is because the model is memorizing the data it has seen and is unable to generalize to unseen examples.



Supervised learning-Linear Regression

- Regularization is a form of regression, that constrains/ regularizes or shrinks the coefficient estimates towards zero. In other words, ***this technique discourages learning a more complex or flexible model, so as to avoid the risk of overfitting.***
- There are multiple different forms of constraints that we could use to regularize. The three most popular ones are Ridge Regression, Lasso, and Elastic Net.

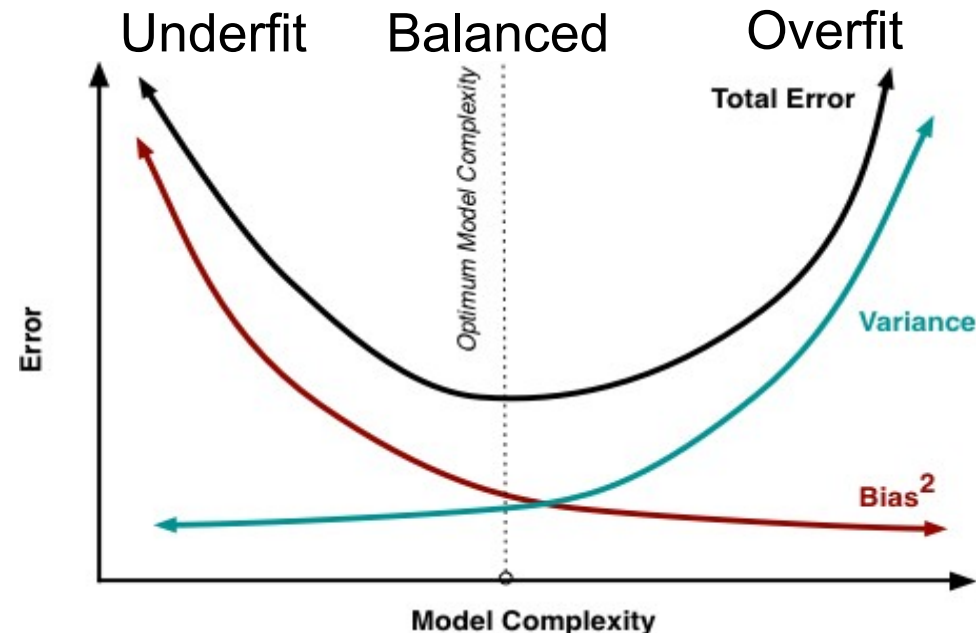


Supervised learning-Linear Regression

Bias and Variance

- Bias occurs when algo has limited flexibility to learn the true signal form a dataset
- Variance refers to an algo's sensitivity to specific sets of training data

Bias is the same as the mean square error (MSE). **Variance** shows how subject the model is to outliers, meaning those values that are far away from the **mean**.



Supervised learning-Linear Regression

Linear Regression Application

Predict Employee Salary based on Years of Experience

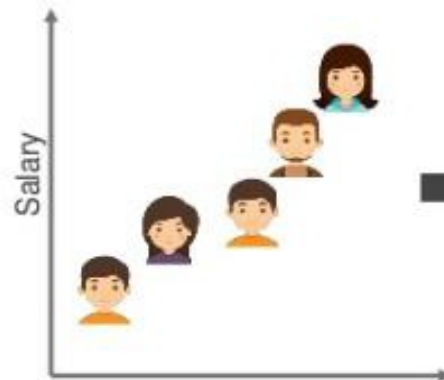
	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0



HR team trying to figure out how much to pay to a new joinee?


New Joinee

How much do we pay her?



Employee data of Salary based on Years of experience

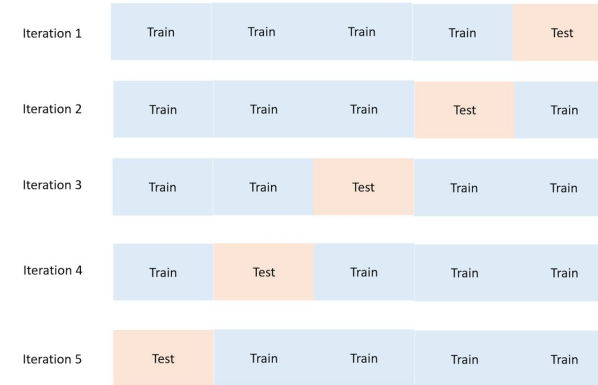


Predict Salary

How to prevent overfitting?

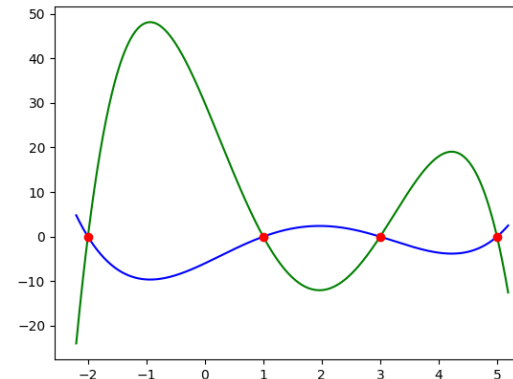
Cross- Validation

Use your initial training data to generate multiple mini train-test splits to tune your model



Regularization

Constraining a model to make it simpler and reduce the risk of overfitting.



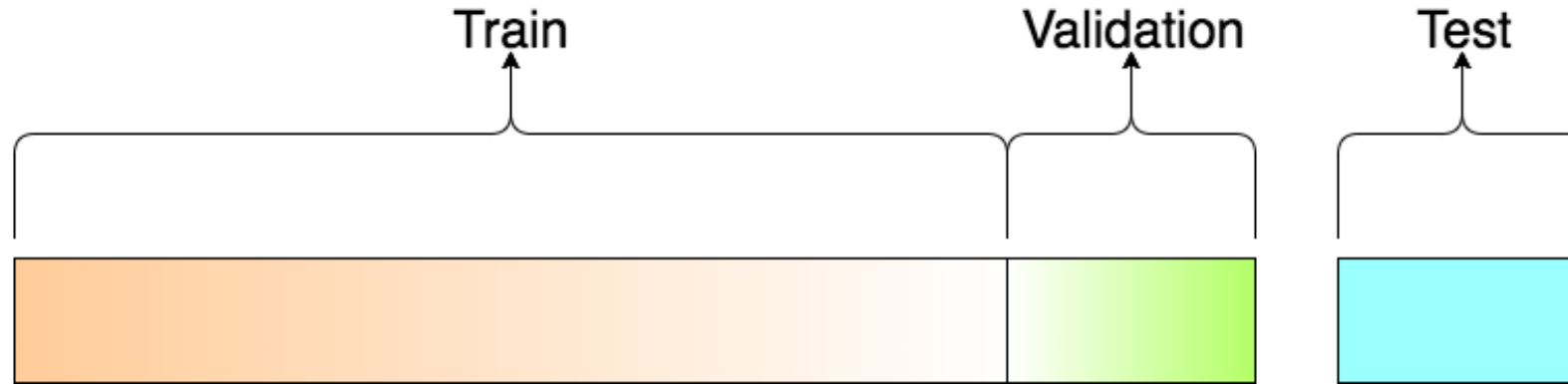
Gather more training data, fix data errors and remove outliers

How to prevent underfitting?

- Increase the size or number of parameters in the model.
- Increase the complexity or type of the model.
- Increasing the training time until cost function is minimised.

Example: Converting a linear model's data into non-linear data. In this case, the transformation of the model leads to it being more unpredictable with respect to any new as well as training data.

Testing & Validation



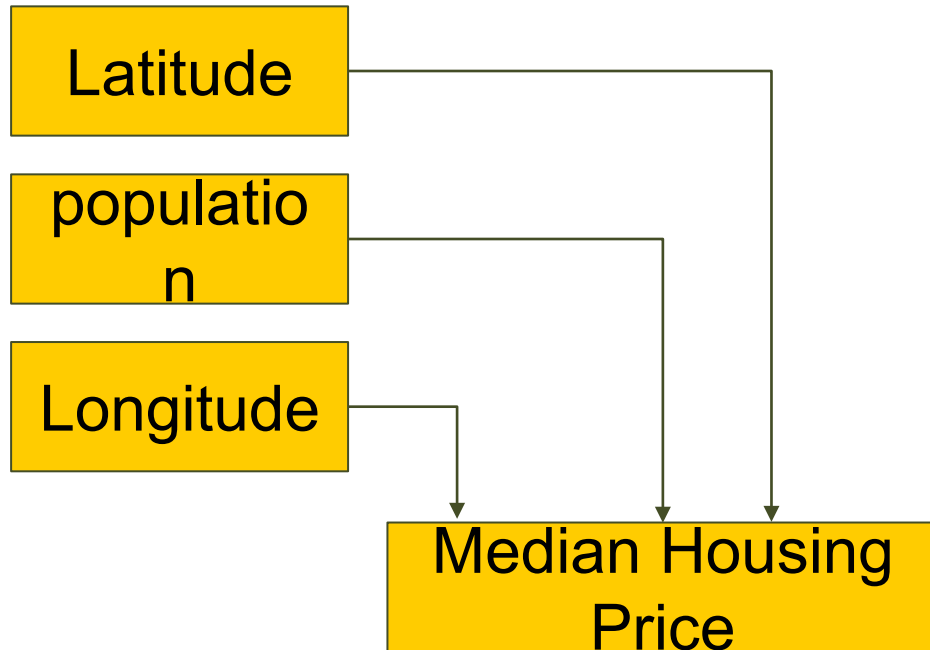
Training Dataset: The sample of data used to fit the model.

Validation Dataset: The sample of data used to provide an unbiased evaluation of a model fit on the training dataset while tuning model hyperparameters.

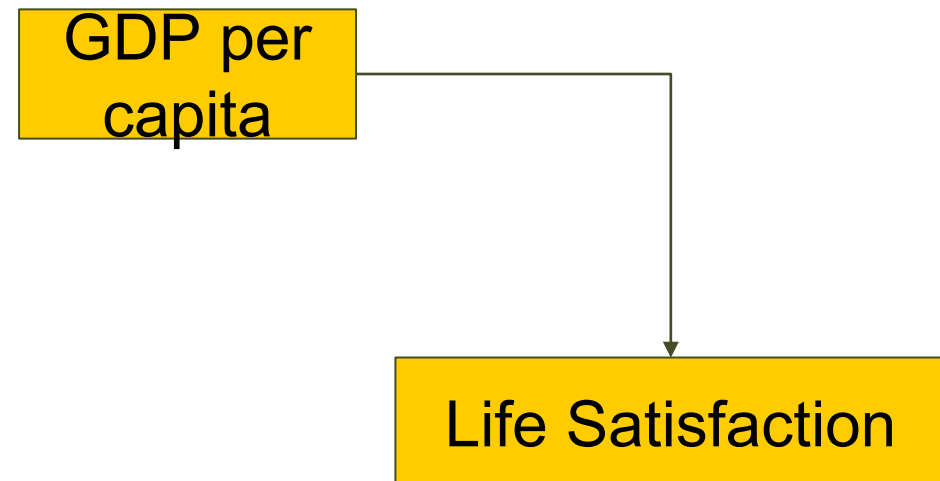
Test Dataset: The sample of data used to provide an unbiased evaluation of a final model fit on the training dataset.

Multiple Linear Regression versus Linear Regression

Multiple Linear Regression



Linear Regression



Multivariate Regression versus Univariate Regression

Multivariate Regression : estimates regression model with more than one outcome variable

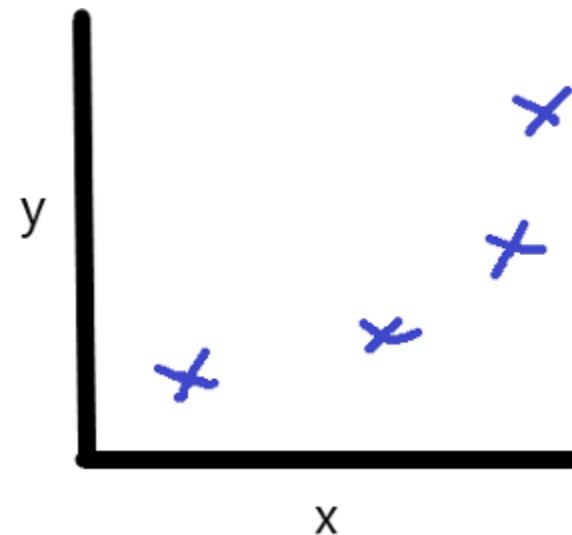
- predict the **directors, music directors** based on the movie budget, duration of movie etc

Univariate Regression : estimates regression model with only one outcome variable

- predict the **Life Satisfaction index** based on the GDP per capita income

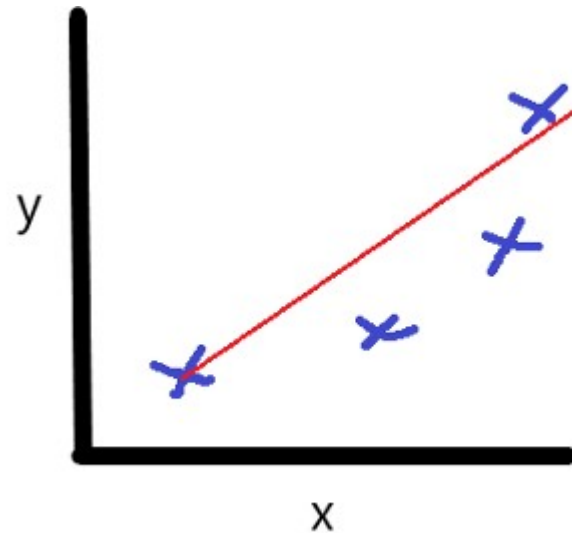
Supervised learning- Polynomial Regression

- Polynomial Regression is another type of regression algorithm in which the relationship between independent and dependent variables is modeled as the nth degree of a polynomial. The polynomial regression fits the non-linear relationship between the independent variable, x, and the dependent variable, y.
- $h(x) = Q_0 + Q_1 * x + Q_2 * x^2 + \dots Q_n * x^n$



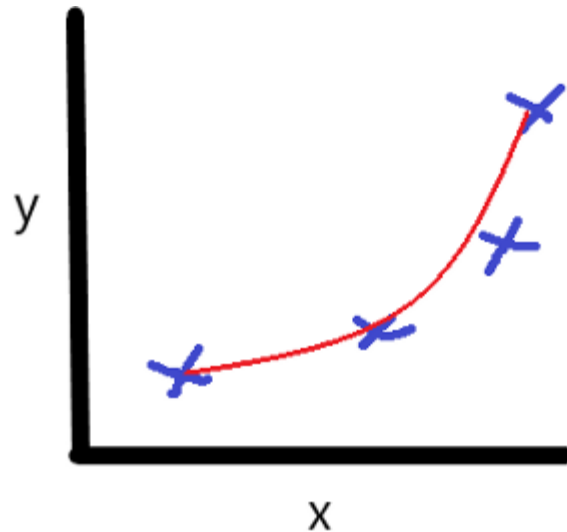
Supervised learning- Polynomial Regression

- We can see that Simple Linear Regression failed to provide the best fit line for the data since the data points are quite far away from the line and the error rate is quite high.



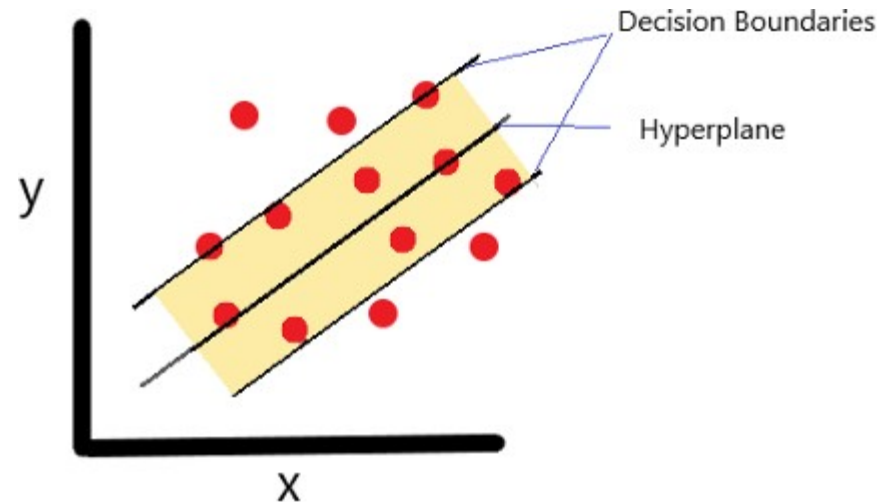
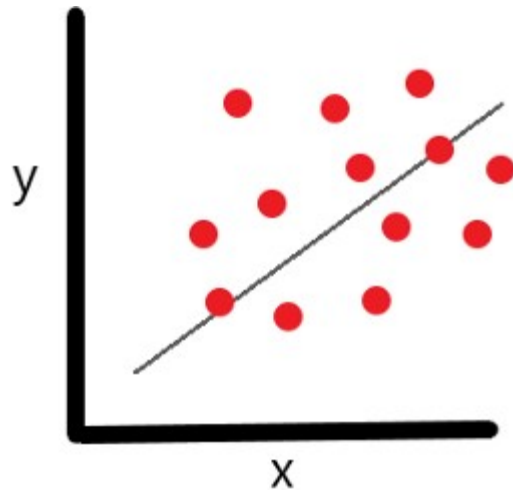
Supervised learning-a Polynomial Regression

- The line (or curve) obtained using polynomial regression is quite optimum since it fits almost all the data points and the error rate is at its minimum. This is why when the two variables have a non-linear relationship, we use the Polynomial Regression algorithm.



Supervised learning- Support Vector Regression(SVR)

- In SVR, we form two lines on either side of the given line, and the data points lying within it are disregarded from the error point of view. The lines around the best fit line are called decision boundaries and the best fit line is called a hyperplane.

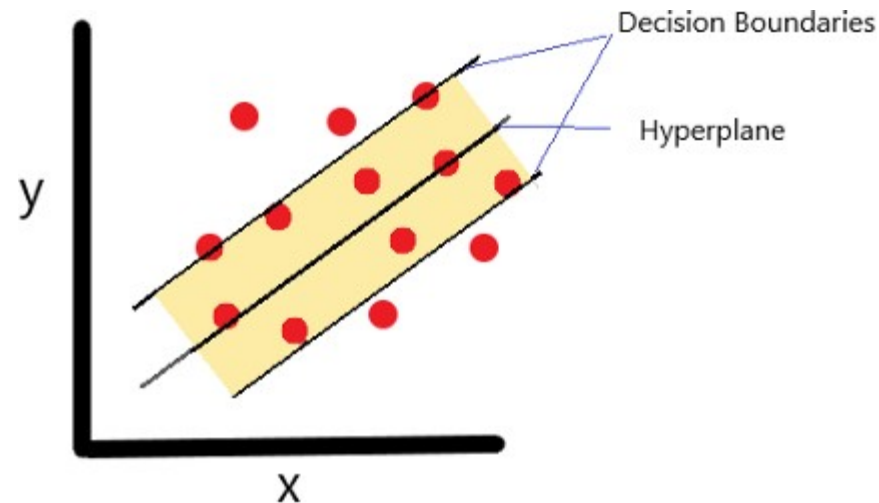


Supervised learning- Support Vector Regression(SVR)

- Our main aim here is to decide a decision boundary at 'a' distance from the original hyperplane such that data points closest to the hyperplane or the support vectors are within that boundary line.*

$Y = wx + b$ (equation of hyperplane)

Then the equations of decision boundary become:
 $wx + b = +a$ $wx + b = -a$



Supervised learning- Support Vector Regression(SVR)

- SVR important parameters are:
 - **kernel**: the kernel type to be used. The most common kernels are **rbf - Radial Basis Function**(this is the default value), **poly** or **sigmoid**, but you can also create your own kernel.
 - **C**: this is the **regularization parameter** described in the section(linear regression)
 - **gamma**: Gamma decides that how much curvature we want in a decision boundary.
- SVR is robust to the outliers.